Reporting Platform

Background:

A nonprofit organization is struggling to track its transactions, making it difficult to clearly see how much money is coming in, how much is going out, and what funds remain available; without this visibility, the organization cannot confidently decide whether it should cut operational costs or reach out for additional funding to sustain its programs. My role was to restructure their raw financial data into a clean staging layer and a transformed layer for their analysis.

Functional Requirement

The customer wants to track the outcomes of the following use cases to help them run the organization more efficiently:

1. **Sponsor Transaction Capture & Monthly Contribution Reporting**

FR-1.1: Daily Sponsor Transaction Ingestion

The system must ingest sponsor transaction data **daily** from source files (CSV, API, or manual upload).

FR-1.2: Data Validation

The system must validate each transaction for:

- Sponsor ID
- Transaction date
- Amount
- Program/campaign code

FR-1.3: Transaction Storage

The system must store validated transactions in the **core layer** and maintain historical records.

FR-1.4: Monthly Contribution Aggregation

The system must automatically aggregate sponsor contributions **monthly** in reporting layer to produce:

- Total current sponsor contributions
- Financial year-to-date contributions
- Total number of sponsors

- Number of sponsors year-to-date

FR-5: Sponsor Classification

The system must classify sponsors as:

- Active
- Repeat
- New
- Inactive

FR-6: Growth Rate Calculation

The system must calculate month-over-month and year-to-date **growth rates**.

FR-7: Contribution by Sponsor Type - Q

The system must generate contribution summaries by sponsor type:

- Individual
- Corporate
- Government
- Community groups

2. Financial Utilization Reporting

FR-2.1: Program & Campaign Data Integration

The system must integrate program/campaign data including:

- Program name
- Start/end dates
- Activities conducted

FR-2.2: Sponsor Onboarding Count

The system must calculate the number of sponsors onboarded per program.

FR-2.3: Program Contribution Summary

The system must aggregate:

- Total sponsor contributions per program
- Total expenses incurred

FR-2.4: Financial Utilization Chart

The system must generate a **Financial Utilization Chart** showing:

- Funds received
- Funds spent
- Remaining balance
- Utilization percentage

FR-2.5: Drill-Down Capability

Users must be able to drill down into:

- Program level
- Campaign level
- Sponsor level

3. Beneficiary Metrics Reporting

FR-3.1: Beneficiary Data Capture

The system must capture beneficiary details including:

- Age
- Gender
- Location
- Type of support received

FR-3.2: Beneficiary Data Validation

The system must validate mandatory fields and flag missing or inconsistent data.

FR-3.3: Beneficiary Demographic Aggregation

The system must aggregate beneficiaries by:

- Age groups
- Gender
- Location

FR-3.4: Support Type Reporting

The system must generate reports showing:

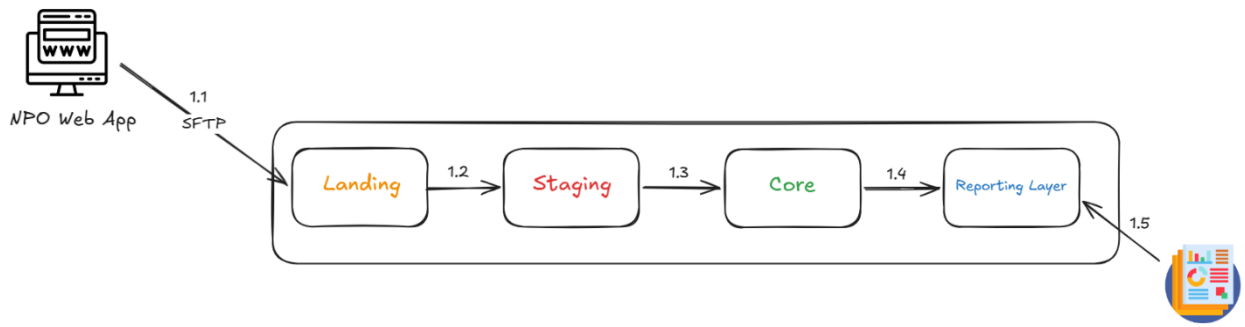
- Number of beneficiaries per support type
- Frequency of support
- Program-wise beneficiary distribution

FR-3.5: Beneficiary Metrics Dashboard

The system must provide a dashboard summarizing:

- Total beneficiaries
- Demographic breakdown
- Support categories

Data Pipeline - End to End Solution Component:



Data Pipeline Stages

Stage 1.1: The NPO Web App captures sponsor registration, sponsor transfer, and beneficiary information through the user interface provided by the frontend. The NPO system generates pipe-delimited CSV files through its extraction process and transfers them to the downstream Reporting Platform via a secure SFTP process.

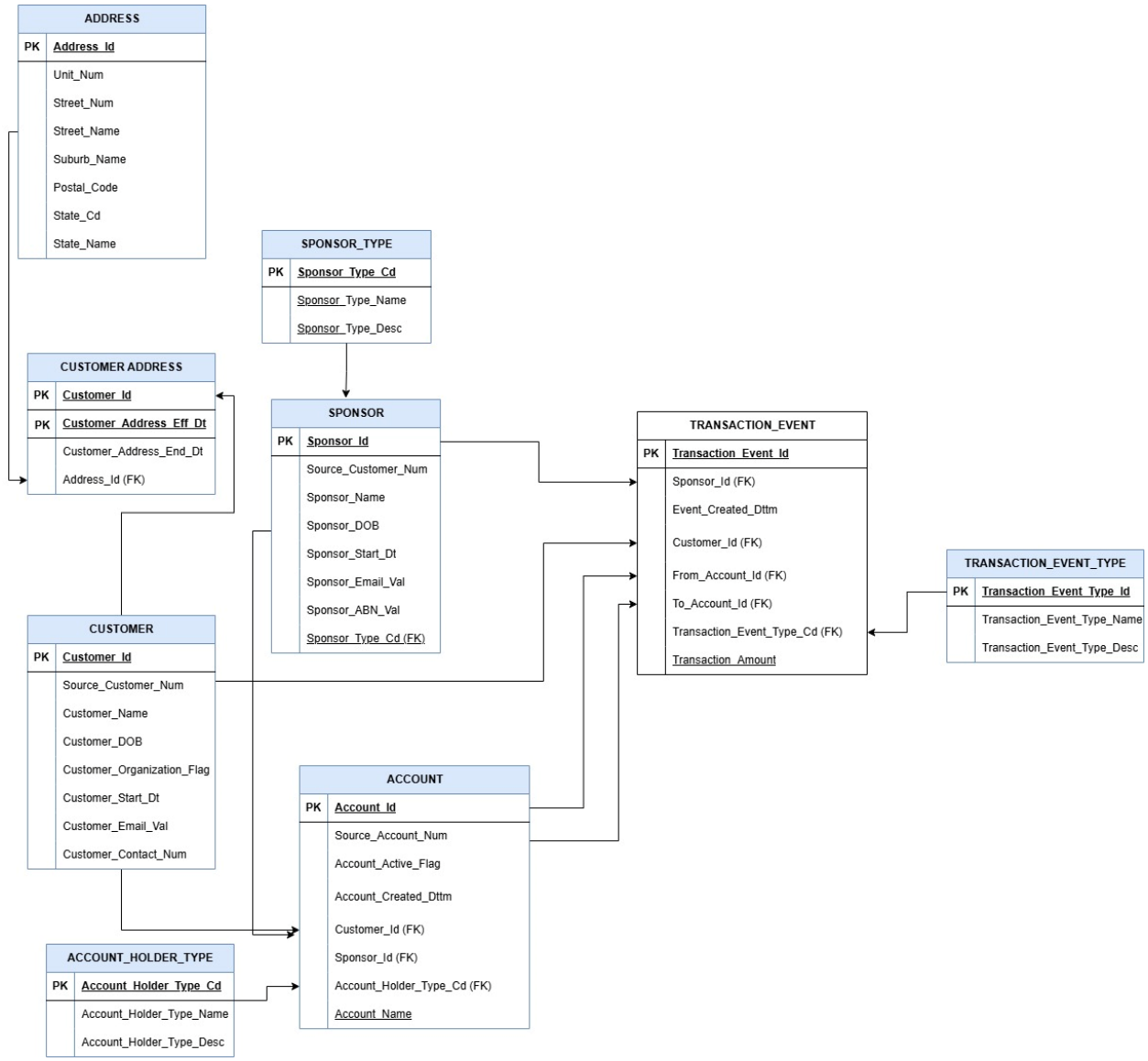
Stage 1.2: Each file is loaded into its corresponding staging table through a one-to-one Python ingestion process. The file name contains the business date, which is added as a column in the staging table to indicate the 'as-on' date of the data. If the load fails, the process triggers notify_logger to send the appropriate message to the support team.

Stage 1.3: The staging tables (such as sponsor, account, address, fund transfer, and beneficiary) are consolidated and transformed into a normalized structure within the core layer.

Stage 1.4: The views are designed to read data from the core layer and fetch the relevant information required for the reporting layer.

Stage 1.5: End users generate reports from the presentation-layer views using their preferred reporting tools. The selection of the reporting tool is currently being finalized

Target DB Structure:



Source to Target Data Mapping:

Version	Source Table	Source Field	Target Table	Target Field	Transformation Logic
1	stg_sponsor		Sponsor	Sponsor_Id	Compare source_registrd_number against Source_sponsor_Num, if not exist, create auto increment number else update the non key fields.
1	stg_sponsor	sponsor_registred_number	Sponsor	Source_sponsor_Num	Direct Mapping
1	stg_sponsor	sponsor_name	Sponsor	Sponsor_Name	Direct Mapping
1	stg_sponsor	sponsor_dob	Sponsor	Sponsor_DOB	Date Validation
1	stg_sponsor	sponsor_start_dt	Sponsor	Sponsor_Start_Dt	Direct Mapping
1	stg_sponsor	sponsor_email_val	Sponsor	Sponsor_Email_Val	Direct Mapping
1	stg_sponsor	sponsor_abn_val	Sponsor	Sponsor_ABN_Val	Direct Mapping
1	stg_sponsor	sponsor_type	Sponsor	Sponsor Type Cd	If sponsor_type = "i" then 1 Else if sponsor_type = "o" then 2 Else sponsor_type = "g" then 3 **refer reference data section below
1	stg_sponsor	contact_number	Sponsor	Sponsor_Mobile_Num	Mobile Num Validation
1			Account	Account id	Compare bank_account_bsb and bank_account_number against Source_Account_Num, if not exist, create auto increment number else update the non key fields.
1	stg_sponsor	bank_account_bsb bank_account_number	Account	Source_Account_Num	Concernate both and map it to target
1			Account	Acccount_Active_Flag	Default to "y"
1	stg_sponsor	businessdate	Account	Account_Created_Dttm	Date Validation
1			Account	Beneficiary_id	Assign null
1	stg_sponsor	sponsor_registred_number	Account	Sponsor_id	Pick up sponsor_id by Look up sponsor. Source_sponsor_Num by sponsor_registred_number
1			Account	Account_Holder_Type_Cd	Assign to 1 because the file is Sponsor file. **refer reference data section below
1			Account	Account Name	--

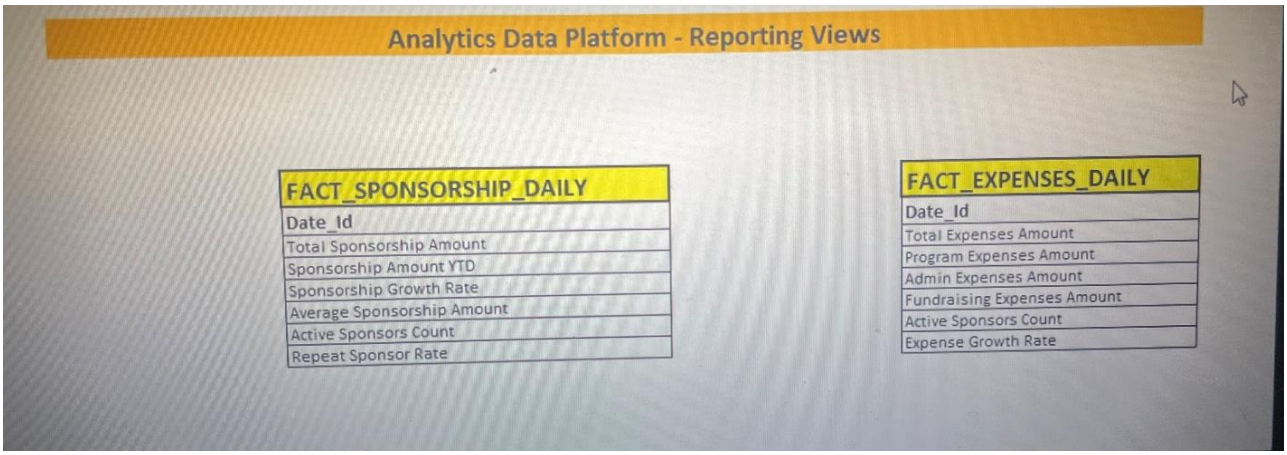
Reference Data:

Sponsor_type

Sponsor type cd	Sponsor type name	Sponsor type desc
1	Individual	Individual sponsor
2	Organisation	Organisation sponsor
3	Government	Government sponsor

Account_holder_type

Account_holder_type_cd	Account_holder_type_ name	Account_holder_type_ desc
1	Sponsor	Account belongs to sponsor
2	Beneficiary	Account belongs to Beneficiary
3	NPO	Account belongs to Non-Profit Organization

Presentation Layer Reporting Views:

The views are designed to read data from the transform layer and extract the relevant information required by the reporting layer.